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A digital workflow proposal for circular economy in building design

Hanne Rangnes Seeberg, Marcin Luczkowski & Sofie Friis Dahl-Nielsen
Structural Engineering Department, NTNU, Norway

ABSTRACT: Transitioning to a sustainable building industry requires incorporating materials from deconstructed buildings into the design of new structures. This paper presents the state of the art in ongoing research on integrating reclaimed load-bearing elements into building design and further discusses how to create a digital workflow that incorporates available reclaimed materials into the conceptual design phase. The research explores the methods and tools necessary for such a digital workflow, focusing on grading and classifying the reclaimed materials, as well as digitally integrating them into the structural design while maintaining the conventional design process. Central to this approach are methods such as 3D scanning, Non-Destructive Testing (NDT), and matching algorithms. The paper discusses current developments, challenges, and future needs for fostering a circular economy in building design, with a focus on reclaimed timber.

1 BACKGROUND

1.1 *Sustainability demands*

In 2022, the building sector contributed 34% of global energy use and 37% of CO2 emissions (United Nations Environment Programme (UNEP) 2024). Initiatives like the European Green Deal aim for net-zero emissions by 2050 (European Commission 2019), supported by the Circular Economy Action Plan, which promotes circular principles throughout building lifecycles (European Commission 2020). Reducing construction and demolition waste requires integrating reclaimed materials into new designs while considering future reuse. This shift introduces challenges for structural engineers as the industry transitions to circular practices.

1.2 *Reclaimed timber*

Reusing building materials from deconstructed structures is one of the most effective strategies for sustainable design (Fivet and Br¨utting 2020). Reclaiming materials minimizes the need for processing, making it preferable to recycling or remanufacturing (Potting, Hekkert, Worrell, and Hanemaaijer 2017). In structural engineering, reuse reduces environmental impact while preserving material integrity (Seeberg, Haakonsen, and Luczkowski 2024). For instance, reclaimed timber not only lowers the demand for new wood but also reduces landfill waste and cuts carbon emissions from harvesting and transportation. This approach aligns with circular economy principles, which emphasize extending material lifecycles through reuse and recycling (Smith 2013), (Davis 2012), (Haasis). Additionally, timber from historical structures can offer superior mechanical properties due to the dense growth of old forests (Davis 2012).

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1.3 *Digital Tools*

The construction industry typically follows a linear model where new materials are chosen as the design progresses, often without considering the reuse of valuable materials from deconstructed buildings. Shifting to circular economy (CE) principles requires significant adjustments to conventional design methods. However, advances in digital technologies are creating new possibilities for sustainable design, with tools like Building Information Modeling (BIM), artificial intelligence (AI), parametric modeling, automation, knowledge-based design, and digital fabrication (Ellen MacArthur Foundation 2019), (Manzoor, Othman, and Pomares 2021)2021)(Zawada,Rybak-Niedzi´ołka, Donderewicz, and Starzyk 2024).

BIM combined with AI has the potential to transform design by optimizing workflows, automating tasks, and enhancing decision-making through predictive models. Generative tools (with AI potential application), such as Spacemaker (Spacemaker AI 2024), Linkio (Linkio 2024), and Spazio (Spazio 2024), enable the generation of multiple design alternatives that support more sustainable choices. Additionally, Algorithm-Aided Design (AAD) and parametric modeling integrate computational algorithms to automate complex tasks, allowing designers to rapidly generate and evaluate alternatives based on structural integrity, material efficiency, and environmental impact.

Moreover, these digital tools can integrate industrial robots and construction machinery within the same digital environment, enabling robotic manufacturing and advanced digital fabrication. This integration supports complex design, automates repetitive and labor-intensive tasks, improves precision, reduces construction time, and enhances safety (Manzoor, Othman, and Pomares 2021)(Zawada, Rybak-Niedzi´ołka, Donderewicz, and Starzyk 2024).

This paper further explores the creation of a digital workflow for reusing reclaimed timber in structural engineering, discussing the essential digital tools, challenges, and areas for future research.

2 METHODS AND TOOLS FOR RECLAIMED MATERIAL INTEGRATION

2.1 *Visual grading*

Visual grading traditionally assesses timber quality by inspecting visible defects such as knots, grain deviations, cracks, and distortions. Trained professionals manually classify timber into strength categories based on these characteristics, following standards like ISO 9709 (International Organization for Standardization 2018), EN 14081-1 (European Committee for Standardization 2019), and NS-INSTA 142 (Standards Norway 2010). Standards specific to reclaimed timber, such as prNS 3691 (Treindustrien 2024), are under development.

Despite its widespread use, visual grading has several limitations, including human error, inability to detect internal defects, time inefficiency, and limited applicability to reclaimed timber. Automated machine grading systems, such as optical scanning and machine learning (ML), have emerged to improve accuracy and consistency by assessing properties like knots and grain orientation, which affect structural performance (Sandvik, Futsather, Liland, and Tomic 2024).

2.2 *Scanning*

3D scanning is the process of digitally capturing an object's precise shape, size, and surface details by mapping it in three-dimensional space. Digital representations from 3D scanning can be crucial for deformation measurement, quality assessment, and ensuring accurate representation of reclaimed elements for design purposes (Hain, Zaghi, Kamali, Zaffetti, Overturf, and Pereira 2019), (Wang, Gao, Cui, Huang, Wu, Mao, and Zheng 2024), (Wondolowski, Hain, and Motaref 2024). These 3D models support circular building design by enabling the integration of reclaimed materials as digital objects into new projects (Yu and Fingrut 2022), (Haakonsen, Tomczak, Izumi,and Luczkowski 2024).

2.2.1 *Scanning methods*

There are various 3D scanning methods, each with unique strengths and applications. For example, photogrammetry creates 3D models by capturing overlapping photos from different angles, and assembling them into a digital representation. This cost-effective and portable method, while less accurate than other techniques, is useful for surveying large stocks of reclaimed wood (Yu and Fingrut 2022), (Wu, Yuan, Tang, and Tian 2022).

Another scanning method is Structured Light Scanning (SLS), which projects a light pattern onto an object's surface, capturing its geometry through distortions in the pattern. This method, widely used in various industries, enables highly accurate 3D models and is effective for structural assessment and cataloging of reclaimed elements in material banks (Hain, Zaghi, Kamali, Zaffetti,Overturf, and Pereira 2019), (Wang, Gao, Cui, Huang,Wu, Mao, and Zheng 2024). Both handheld scanners and stationary cameras can perform SLS, offering flexibility depending on the object's size and shape (Hain, Zaghi, Kamali, Zaffetti, Overturf, and Pereira 2019), (Gundersen and Dahl-Nielsen 2024).

Finally, LiDAR (Light Detection and Ranging) uses laser beams to measure the time light takes to travel to an object and back, creating a point cloud. LiDAR scanning can be performed using two main techniques: time-of-flight (ToF) and phase-shift. ToF measures the travel time of laser pulses, making it suitable for long-range measurements, while phase-shift, which measures the phase difference between emitted and reflected light, provides higher accuracy at shorter ranges (Wang, Tan, and Mei 2020). LiDAR can also be applied in different platforms: Terrestrial Laser Scanning (TLS) is ideal for detailed, stationary scans of buildings and infrastructure, offering high-density data capture with precision. Mobile Laser Scanning (MLS), often mounted on vehicles, is suited for large-scale surveys with slightly lower precision (Wang, Tan, and Mei 2020). Smartphone LiDAR technology, though slower and less accurate, offers a portable solution for small-scale or quick assessments (Gundersen and Dahl-Nielsen 2024).

2.2.2 *Data formats and digital objects*

3D scanning typically produces models in two formats: meshes and point clouds. Meshes represent geometry through interconnected vertices (usually polygons), while point clouds consist of discrete points that represent spatial distribution (Robert McNeel and Associates). For design purposes in the AEC industry, these formats can be converted into digital objects called Boundary Representation (BRep), which is essential for compatibility with Building Information Modeling (BIM) environments. BRep defines models using geometrical equations, making it highly precise and easily convertible to Industry Foundation Classes (IFC), which structure both geometrical data and essential metadata for building objects to ensure compatibility across BIM software (Fu, Peddireddy, Zhou, Xi, Aggarwal, Li, and Jun 2023). BRep models can also be exported to finite element analysis (FEA) software, enabling seamless integration into both design and analysis processes (Fu, Peddireddy, Zhou, Xi, Aggarwal, Li, and Jun 2023). A master's thesis from NTNU (Gundersen and Dahl-Nielsen 2024) explored structured light scanning (SLS) for reclaiming structural timber, testing various methods for converting point clouds into BRep models and emphasizing the importance of balancing detail with practicality. Aagaard and Larsen (Aagaard and Larsen 2020) developed a similar workflow for sawlogs, combining Grasshopper components and custom Python scripts to generate NURBS polysurfaces, which are smooth, mathematically defined surfaces used to accurately model complex shapes.

Scanning involves both hardware and software components. Many scanners come with proprietary software for aligning scans, noise removal, and generating point clouds or meshes. Noise filtering algorithms are crucial in cleaning point cloud data before conversion to BRep (Wang,Tan, and Mei 2020). Hanocka et al. (Hanocka, Metzer, Giryes, and Cohen-Or 2020) propose a more automated approach using a convolutional neural network (CNN)-based self-prior, simplifying the conversion process from point clouds to meshes, which are more manageable for digital modeling and enhance the integration of scanned objects into design workflows.

2.3 *Non-Destructive Testing (NDT)*

NDT is essential for classifying reclaimed timber without damage, providing precise internal measurements that ensure key structural properties like modulus of elasticity (MOE) for stiffness and modulus of rupture (MOR) for strength– both vital for structural design.

2.3.1 *Ultrasound Testing*

Ultrasound testing estimates timber stiffness and detects internal defects by measuring the velocity of ultrasound waves traveling through the tested element. Based on these velocity measurements, the dynamic MOE is calculated, incorporating density as an input. Unlike the static MOE used in design, the dynamic MOE requires adjustments through regression models to accurately predict static MOE and MOR, which better reflect structural behavior. These regression models also consider parameters such as knots, grain slope, density, and moisture content to ensure reliable predictions (Arriaga, Osuna-Sequera, Bobadilla, and Esteban 2022).

X-ray and CT scanning provide detailed internal visualizations of timber, measuring density and moisture content– both critical inputs for accurate ultrasound predictions. By analyzing X-ray absorption (CT numbers), density can be estimated precisely, with moisture content affecting absorption rates. Although accurate, CT scanning is slower and requires significant expertise (Jaskowska-Lema´nska and Przesmycka 2021).

Computer vision, which uses image processing and ML techniques to interpret visual data, automates the detection of knots and grain orientation, improving the grading process (Daval, Pot, Belkacemi, Meriaudeau, Collet, and Collet Au 2015) (Sejersted and Viken 2024). High-resolution images are analyzed using algorithms to measure grain slope and knot characteristics, crucial for accurate MOR and MOE predictions. Sejersted and Viken combined computer vision with ultrasound and regression models to enhance predictions for reclaimed timber properties.

2.3.2 *Lamb Wave, Vibration Testing, Stress Wave*

Fathi et al. introduced Lamb wave propagation as an innovative NDT method that, when combined with ML, estimates MOE and MOR by measuring Lamb wave velocity through timber (Fathi,Nasir, and Kazemirad 2020). These waves interact with the internal structure, making this method highly effective for detecting small defects. A neural network then processes data on Lamb wave velocity, density, and moisture content to accurately predict MOE and MOR. However, Lamb wave propagation is best suited for thin plates.

Vibration testing, another NDT technique, induces vibrations and analyzes the resulting longitudinal and transverse frequencies in the timber to estimate its dynamic MOE, providing insights into the timber's overall mechanical properties. This method shows a strong correlation with static MOE, though its precision in detecting smaller local defects is limited compared to ultrasound (Acu˜na, Mart´ınez L´opez, Spavento, Casado, Alvarez-Martinez, O'Ceallaigh, Harte, and Balmori 2023).

Stress wave propagation is another NDT method that uses lower-frequency mechanical waves than ultrasound to assess timber's stiffness and detect large structural anomalies, such as decay and major cracks. While it provides reliable evaluations of global properties, its resolution for detecting smaller defects is lower compared to ultrasound. Stress wave velocity, combined with data on density and moisture content, is used to estimate dynamic MOE (Dackermann, Crews,Kasal, Li, Riggio, Rinn, and Tannert 2014).

2.3.3 *Reclaimed Timber Grading*

Combining NDT, ML, and computer vision enhances timber grading and can support or even substitute traditional visual grading. ML models trained on data from visual parameters (such as knots and grain slope) and NDT methods (e.g., ultrasound, vibration testing) have the potential to accurately predict MOE and MOR without destructive testing, enabling automated and precise classification of reclaimed timber elements (Acu˜na-Rello, Spavento, Casado-Sanz, Basterra,L´opez-Rodr´ıguez, Ram´on-Cueto, Relea-Gangas, Morillas-Romero, Escolano-Margarit, Mart´ınez,and Balmori 2022), (Sejersted and Viken 2024).

2.4 *Matching Algorithms for Structural Design*

Matching algorithms play a critical role in structural engineering by optimizing the reuse of reclaimed materials. These algorithms help select and allocate elements from available stocks, minimizing waste and ensuring efficient material usage. Common algorithms include greedy methods, genetic algorithms, and bipartite graphs, which vary in computational complexity and performance (Mollica and Self 2016), (Brutting, Ohlbrock, Hofer, and Dacunto 2021), (Warmuth,Estrella, K¨alin, Natterer, and Fivet 2023).

Dynamic relaxation and growth algorithms are employed to adjust structural geometry when integrating irregular or reclaimed materials, ensuring the design remains stable and structurally sound. These tools work in conjunction with matching algorithms to optimize material placement (van Lookeren Campagne, Sonneveld, van der Meer, Noteboom, and Kavoura 2023), (Lokhandwala 2018).

In a top-down approach, reclaimed materials are matched to a predefined design, optimizing their use within the intended structure. The bottom-up approach, by contrast, starts with the available stock of reclaimed materials and designs the structure around these elements. Both methods rely on matching algorithms for efficient integration of materials (Tomczak, Haakonsen, and Luczkowski 2023), (Philippe Block, Boller, DeWolf, Pauli, Kaufmann, Dupas, and Matthias Hudert 2024).

3 DISCUSSION

The transition toward sustainable building practices, particularly the reuse of materials, requires an integrated workflow combining material testing and computational design tools. This workflow bridges the gap between the physical properties of reclaimed elements and their seamless integration into modern designs. It consists of two phases: creating a material dataset and integrating these materials into a conceptual digital design.

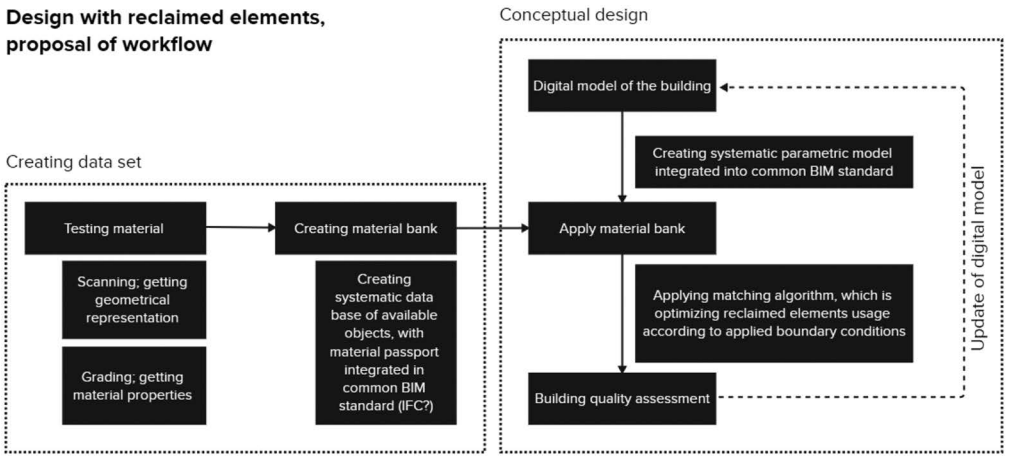


Figure 1. A digital workflow for circular economy in building design.

3.1 *Creation of the Material Dataset*

This phase focuses on collecting, testing, and digitally representing reclaimed materials. Structural materials, such as timber, are subjected to various NDT methods to assess their geometric and mechanical properties. Scanning techniques like LiDAR and photogrammetry generate high-resolution digital models, capturing irregularities in reclaimed materials. These models are

typically converted to Boundary Representation (BRep) formats compatible with CAD and BIM software.

NDT methods such as ultrasound, vibration testing, stress wave propagation, lamb wave propagation, and X-ray/CT scanning are used to measure structural properties like the MOE and MOR. Additionally, due to the variability in reclaimed materials, accurate grading is essential, and techniques like computer vision, ML, and regression models are being explored to identify defects and enhance prediction accuracy.

The scanned and tested materials are then organized into a material bank, a digital repository cataloging geometric and mechanical properties. Each material is assigned a "material passport" with relevant information like size, strength, and condition. This material bank integrates seamlessly with BIM standards (e.g., Industry Foundation Classes, IFC), enabling easy incorporation of reclaimed materials into modern architectural and engineering projects.

3.2 *Conceptual Design and Integration of Reclaimed Materials*

Once the material bank is established, the next phase of the workflow focuses on applying these reclaimed elements to a conceptual design. This process begins with the creation of a digital parametric model of the building. The parametric model serves as a flexible template for the building's structure, and allows for the rapid adjustment of design elements based on constraints such as load-bearing requirements, aesthetic considerations, and, importantly, the availability of reclaimed materials.

To incorporate reclaimed elements, the workflow applies the material bank to the digital parametric model through an optimization process driven by computational algorithms. This involves using matching algorithms, matching the material properties and geometries from the material passport of reclaimed elements with the requirements of the structural design.

Further, the matched reclaimed elements are integrated into the design, and the digital model is iteratively updated, ensuring structural soundness while optimizing material reuse. This feedback-driven process ensures a balance between sustainability and engineering efficiency.

After material matching and integration, the updated digital model undergoes a quality assessment to ensure that the design, now incorporating reclaimed materials, meets safety and performance standards. Earlier discussed NDT, enhanced with AI, provides a more accurate structural assessment of materials, verifying their suitability for load-bearing roles. During the quality assessment phase, any adjustments made from material integration are reviewed to ensure structural integrity and compliance with building codes, fostering greater confidence in reclaimed materials for mainstream construction applications.

3.3 *Challenges and Future Considerations*

While the discussed workflow provides a clear structure for integrating reclaimed materials into building design, several challenges remain. One of the most pressing issues is the lack of robust building assessment protocols, particularly when it comes to using reclaimed materials. Current assessment frameworks are often vague, lacking standardized metrics to evaluate buildings that incorporate reused elements. This "mistiness" in building assessment protocols complicates the quality assurance process and creates uncertainty around compliance with building codes. Developing clear, quantifiable metrics for assessing the structural integrity, safety, and environmental impact of buildings with reclaimed components will be critical to future advancements in this area.

In addition, ensuring compatibility between digital models generated through scanning and existing Building Information Modeling (BIM) systems remains a significant technical challenge. While Boundary Representation (BRep) models offer a useful format, converting scanned data into this format without losing precision or creating errors is still an evolving area that requires further research and development.

Another issue is the grading of reclaimed materials, where improvements in predictive accuracy are needed. Although ML is a promising tool for enhancing grading, the variability of reclaimed elements – especially their defects and non-uniformities – makes automation difficult. More

comprehensive datasets and better regression models will be necessary to fine-tune predictions for material performance.

Moreover, computational efficiency in the material matching process needs improvement. Advanced algorithms like genetic and bipartite graph methods are effective but can be computationally expensive, especially in complex designs or large-scale projects. Optimizing these algorithms for speed and accuracy will be crucial for the widespread adoption of this workflow in the construction industry.

Addressing these challenges will require significant innovation in both digital tools and structural engineering methods. By refining assessment protocols, enhancing material grading techniques, and streamlining algorithmic processes, this workflow can play a pivotal role in making reclaimed materials a viable option for mainstream construction. The continued development of clear, reliable metrics for building assessment will also be essential to fostering trust and adoption of these sustainable practices across the industry.

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